

Edukit Rotary Inverted Pendulum: Real Time Control System Workbench

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1. Introduction: Real Time Control System Workbench

The Edukit Rotary Inverted Pendulum may operate independently of any computing platform. Its control system implementation is supported in an open source high speed, real time C code implementation operating on the STM32 processor.

A Real Time Control System Workbench is available that communicates commands and receives data for real time display. This enables:

- 1) Real time configuration of Stepper Motor Interface parameters
- 2) Real time configuration of Control System parameters
- 3) Real time Display of Motor Interface and Control System parameters
- 4) Real time display of user-selected for Sensitivity Function Step Response Measurements
- 5) Selection of Data Logging operation with measured data archived in a user-selected directory

[The Real Time Control System Workbench for Matlab is available for download at this link.](#)

The Real Time Control System Workbench is designed to support both Windows and Mac OSX platforms. This system samples and displays data at a rate of 50 Hz (a rate limit set serial data transfer rate and by the graphics bandwidth of Windows and Mac OSX platforms).

An additional high speed sampling system interface operating on both Windows and Mac OSX is available, for high speed sampling and storage of data at the full 500 Hz rate of the Edukit system. This is accompanied by data processing provided by direct computation and display of system response and each of the four Sensitivity Function frequency response spectra.

Figure 2 shows the user interface of the Real Time Control System Workbench. The user has selected the options of Sine Off (removing a sinusoidal tracking reference command signal) and adding Rotor Step On (providing a step signal occurring each 16,384 control steps and 32 seconds).

2. System Default Mode Operation

System Requirements

- 1) The Edukit system requires a standard USB 2.0 interface provided by a USB 5.0V power source or a standard Windows or Mac OSX computer.
- 2) Your computer requires a USB interface:

Some Mac laptops of the 2015 and previous versions, include USB ports. More recent versions require an adapter from their ports to USB. This is the Apple USB-C to USB Adapter.

3. System Power and Data Interface

- 1) Power is first applied to the IHM01A1 Stepper Motor Controller by attaching the Edukit DC power supply unit to a 110V or 220V AC mains connection.
- 2) The Edukit USB cable is then attached to either a computer or to a USB 5V power supply adapter.

4. System Start

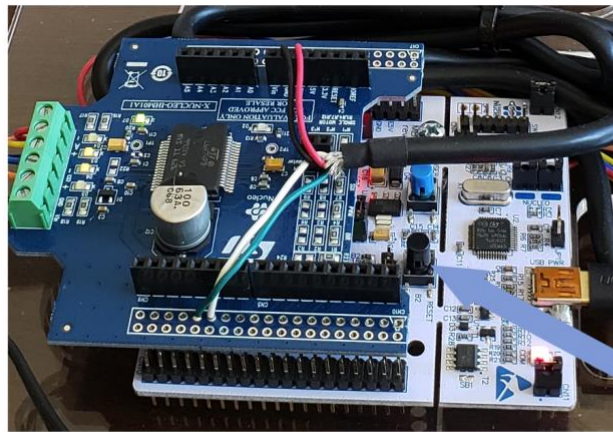
When power is applied to the Edukit system by its USB cable, its processor will boot and launch a process to provide stable inverted pendulum operation. These steps will occur:

- 1) The Edukit system will transmit a prompt to a computer that is connected to the USB cable. The prompt requests entry of a mode selection.
- 2) If no response is returned from the computer after a waiting time period of 10 seconds, a default demonstration mode operation starts.
- 3) The Edukit system will first determine if the pendulum is moving or is motionless in a stable position.
- 4) If the pendulum is moving, the Edukit system will wait until the pendulum is motionless and vertical down.
- 5) If stable, the Edukit system will measure the angle of the pendulum relative to vertical as required for control action to follow.
- 6) Then, when complete, the Edukit system will signal the user by displacing the rotor in a sequence of small motions as an alert signal.
- 7) The user should then raise the pendulum to a vertical position.
- 8) When the pendulum angle is within three degrees of vertical, the control system will start and stable operation will commence.
- 9) After a 10 second period of operation, an amplitude-modulated sinusoidal reference tracking signal will start and inverted pendulum operation will display rotor angle control as well as pendulum control.

- 10) If the pendulum motion vertical positioning does not occur, the Edukit system will wait, perform the angle calibration and signal the user again.

5. System Reset and Restart

The Edukit system may also be restarted by application of the Reset button. This is the black button on the Nucleo board shown in *Figure 1*.



System
Reset

Figure 1. The Edukit system may be reset by depressing this reset button on the Nucleo board.

6. Starting the Real Time Control System Workbench

The Real Time Control System Workbench may be started from Matlab at anytime.

However, if another resource is using the serial interface to Edukit (for example, the screen utility of Mac OSX or putty interface of Windows) then these must be terminated first. These systems are described in the next section.

Graphical Data Windows at Left Column

First, the data displayed and user controls are summarized here.

- 1) Display of selected Sensitivity Function including Complimentary Sensitivity Function, Sensitivity Function, Load Disturbance Sensitivity Function and Noise Rejection Sensitivity Function.
- 2) Rotor Step signal indicator (On or Off)
- 3) Pendulum Controller Proportional Gain
- 4) Pendulum Controller Integral Gain
- 5) Pendulum Controller Derivative Gain
- 6) Rotor Controller Proportional Gain
- 7) Rotor Controller Integral Gain
- 8) Rotor Controller Derivative Gain
- 9) Stepper Motor Torque Current
- 10) Stepper Motor Maximum Acceleration (applicable during Motor Test operation)
- 11) Stepper Motor Maximum Deceleration (applicable during Motor Test operation)
- 12) Stepper Motor Maximum Speed (applicable during Motor Test operation)
- 13) Stepper Motor Maximum Speed (applicable during Motor Test operation)
- 14) Step Size of parameter value adjustment steps
 - a. This size ranges from values of 0.5, 2, 20, 50, 1 and 100
 - b. This is increased or decreased by clicking on Inc Step or Dec Step buttons

Control System Configuration Control Buttons Lower Row of Interface

- 1) Exit: Exit Real Time Workbench
- 2) Stop Control: Terminate control and exit Real Time Workbench
- 3) Sine On: Enable Modulated Sine Waveform Rotor Tracking Reference
- 4) Sine Off: Disable Modulated Sine Waveform Rotor Tracking Reference
- 5) Inc Step: Increase Step Size of user parameter value input
- 6) Dec Step: Decrease Step Size of user parameter value input

Each of the following controls adjust parameter values by an amount equal to Step Size for each button click

- 7) Inc Pend P: Increase Pendulum Controller Proportional Gain
- 8) Inc Pend P: Decrease Pendulum Controller Proportional Gain
- 9) Inc Pend I: Increase Pendulum Controller Proportional Gain
- 10) Inc Pend I: Decrease Pendulum Controller Proportional Gain
- 11) Inc Pend D: Increase Pendulum Controller Proportional Gain
- 12) Inc Pend D: Decrease Rotor Controller Proportional Gain
- 13) Inc Rotor P: Increase Rotor Controller Proportional Gain
- 14) Inc Rotor P: Decrease Rotor Controller Proportional Gain
- 15) Inc Rotor I: Increase Rotor Controller Proportional Gain
- 16) Inc Rotor I: Decrease Rotor Controller Proportional Gain
- 17) Inc Rotor D: Increase Rotor Controller Proportional Gain
- 18) Inc Rotor D: Decrease Rotor Controller Proportional Gain

1. Control System Configuration Control Buttons Right Column of Interface

- 1) Log Data: Start data logging of three 40 sec sweeps starting at next sweep
- 2) Rotor Step On: Enables Rotor Step reference track signal
- 3) Rotor Step Off: Disables Rotor Step reference track signal

Note also:

- 4) Rotor Step On: Enables Complimentary Sensitivity Function measurement
- 5) Rotor Step Off: Disables Complimentary Sensitivity Function measurement
- 6) Pendulum Impulse On: Enables Pendulum Impulse reference track signal
- 7) Pendulum Impulse Off: Disables Pendulum Impulse reference track signal
- 8) Load Dist On: Enables Load Disturbance Rejection Sensitivity Function measurement
- 9) Load Dist On: Disables Load Disturbance Rejection Sensitivity Function measurement
Note that Load Disturbance response is small for effective design. Thus, this signal is displayed with a 10 times magnification.
- 10) Noise Sens On: Enables Noise Rejection Sensitivity Function measurement
- 11) Noise Sens Off: Disables Noise Rejection Sensitivity Function measurement
- 12) Sensitivity On: Enables Sensitivity Function measurement
- 13) Sensitivity Off: Disables Sensitivity Function measurement

The following controls apply when the system is operating in Motor Control Characterization Mode as selected by the Edukit Serial Interface. These controls are not applicable or active in normal closed loop operation.

- | | |
|------------------|---|
| 14) Dec Min Spd: | Decreases minimum speed in by amount equal to selected Step Size. |
| 15) Inc Min Spd: | Increases minimum speed in by amount equal to selected Step Size. |
| 16) Dec Max Spd: | Decreases maximum speed in by amount equal to selected Step Size. |
| 17) Inc Max Spd: | Increases maximum speed in by amount equal to selected Step Size. |
| 18) Dec Torq: | Decreases Motor Torque Current by amount equal to selected Step Size. |
| 19) Inc Torq: | Increases Motor Torque Current by amount equal to selected Step Size. |

Examples of Real Time Workbench views for multiple modes include these:

- 1) *Figure 2: Rotor Response and Complimentary Sensitivity Function step response*
- 2) *Figure 3: Sensitivity Function step response*
- 3) *Figure 4: Load Disturbance Sensitivity Function step response*
- 4) *Figure 5: Noise Rejection Sensitivity Function step response*

Note that the Real Time Workbench provides effective measurement of the response of each of the four Sensitivity functions.

In addition, as noted, an additional high speed sampling system, the Sensitivity Function Measurement System operating on both Windows and Mac OSX is available, for high speed sampling and storage of data at the full 500 Hz rate of the Edukit system. This is accompanied by data processing provided by direct computation and display of system response and each of the four Sensitivity Function frequency response spectra.

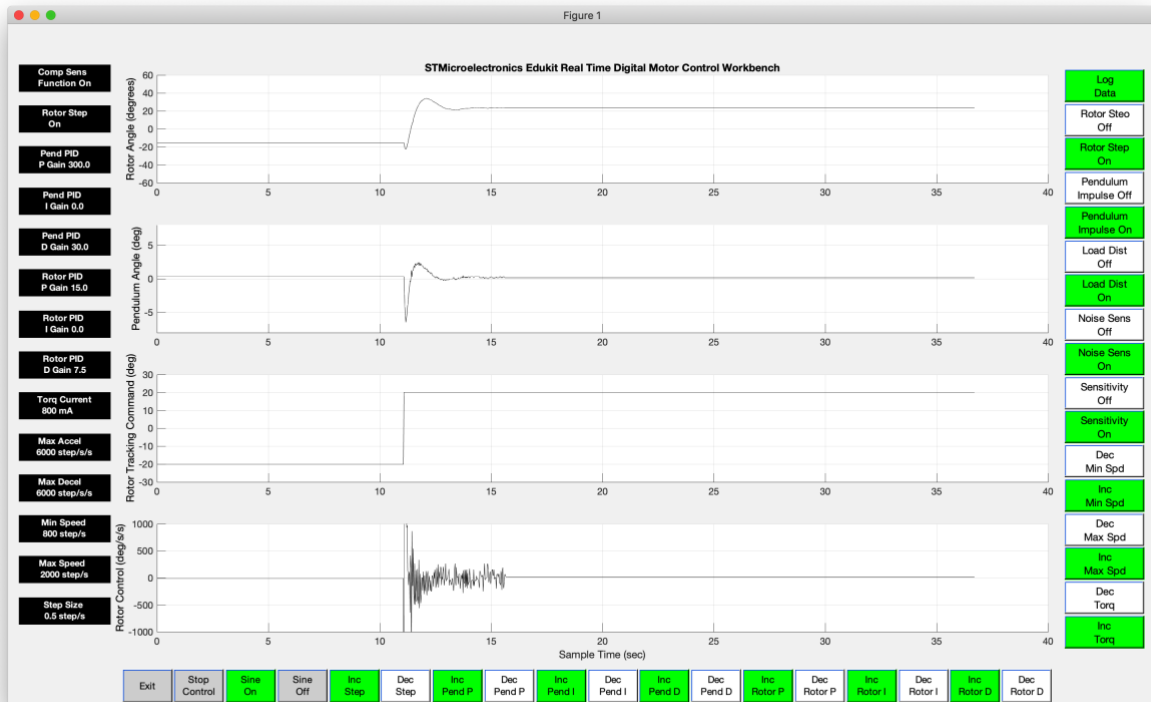


Figure 2. The Edukit Rotary Inverted Pendulum Real Time Control System Workbench in Matlab. This shows part of a trace of real time data corresponding to operation with system response to a 40 degree step in the Rotor Angle Tracking Command. This also provides measurement of the Complimentary Sensitivity Function step response.

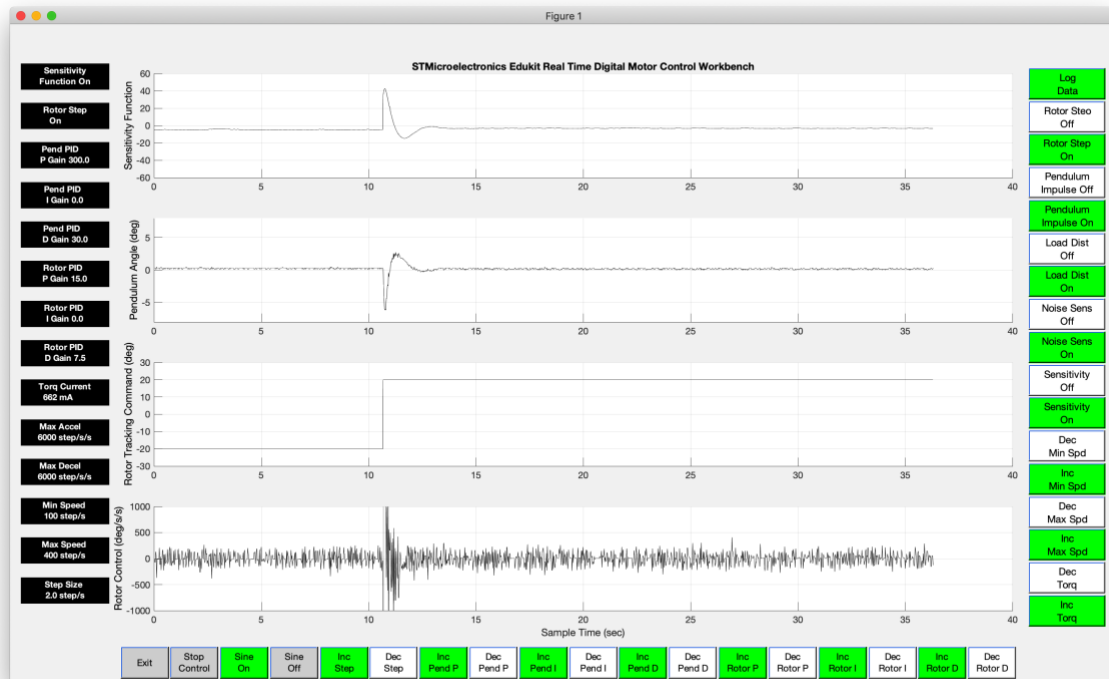


Figure 3. The Edukit Rotary Inverted Pendulum Real Time Control System Workbench. This shows measurement of the Sensitivity Function with part of a trace of real time data.

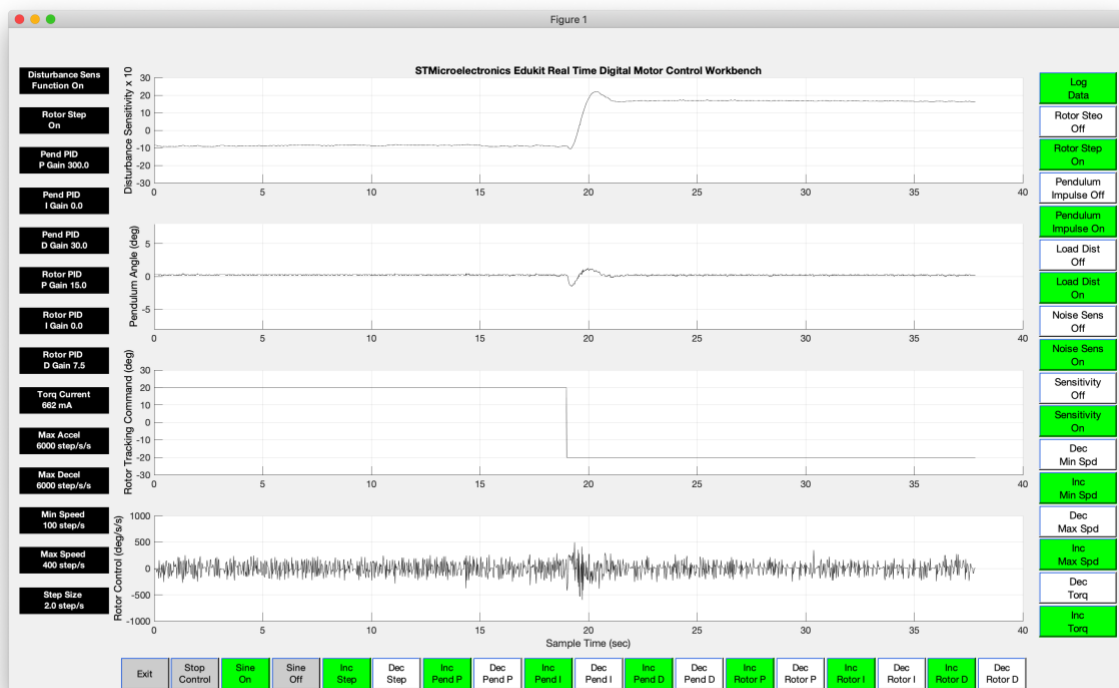


Figure 4. The Edukit Rotary Inverted Pendulum Real Time Control System Workbench. This shows measurement of the Load Disturbance Rejection Sensitivity Function with part of a trace of real time data. Note that Load Disturbance response is small for effective design. Thus, this signal is displayed with a 10 times magnification.

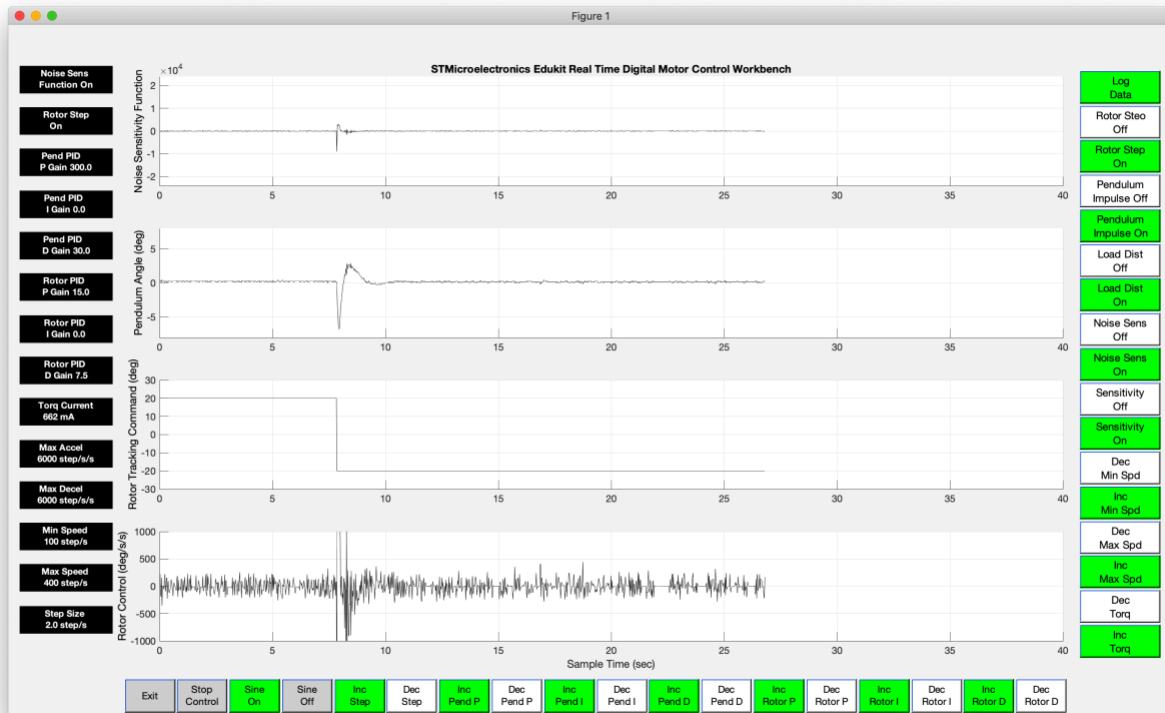


Figure 5. The Edukit Rotary Inverted Pendulum Real Time Control System Workbench. This shows measurement of the Noise Rejection Sensitivity Function with part of a trace of real time data.

7. Edukit Sensitivity Function Frequency Response Computation System

The four Sensitivity Functions that enable performance characterization of control systems are each measured by the Edukit system. These are applied and described with examples in Sections **Error! Reference source not found., Error! Reference source not found., Error! Reference source not found., Error! Reference source not found..**

The Sensitivity Function measurements may be selected by the Control System Workbench using the control buttons of

1. Sensitivity On/Off: Selects Sensitivity Function
2. Rotor Step On/Off: Selects Complimentary Sensitivity Function
3. Load Dist On/Off: Selects Load Disturbance Rejection Sensitivity Function
4. Noise Sens On/Off: Selects Noise Sensitivity Function

Operation

To acquire Sensitivity Function data, follow this procedures

1. First select one of the Sensitivity Function options.
2. Then, select Log Data
3. At this time, select a number of measurement Sweeps. Each sweep requires 40 seconds.
4. The default and minimum for Frequency Response Computation is 3 sweeps. However, this can be increased to improve signal to noise ratio in measurements.

Output data will be stored in *.mat files in the directory selected at the start of Workbench operation. The data files will be of the format:

1. Sensitivity Function
 - a. Edukit_Sensitivity_Function__<Date>_<Time>.mat
2. Complimentary Sensitivity
 - a. Edukit_Step_Response_Rotor_Angle_(degrees)__<Date>_<Time>.mat
3. Load Disturbance Rejection Sensitivity Function
 - a. Edukit_Disturbance_Sensitivity_Function__<Date>_<Time>.mat
4. Selects Noise Sensitivity Function
 - a. Edukit_Noise_Sensitivity_Function__<Date>_<Time>.mat

Frequency Response Computation

The Edukit system includes the capability for introduction of a step signal stimulus for probing of the control system response for each of the four Sensitivity Functions.

Frequency response for each Sensitivity Function may then be determined by computing the ratio of the Fourier Transform of both the system response and the system stimulus for each frequency component.

A Matlab script, [Edukit Sensitivity Function Freq Response Computation.m](#) is provided for this computation. This operates on a selected Sensitivity Function data file and returns the computed frequency response spectrum.

This system will operate on a minimum of three data sweeps. However, as desired, this system will also signal average a greater number of sweeps. This may be useful in computation of frequency response over a very large dynamic range and for small magnitude in the frequency response spectrum.